

UNSTEADY AERODYNAMIC AND AEROACOUSTIC SIMULATIONS OF A FENESTRON TAIL ROTOR

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ABSTRACT

The main scope of this study is the numerical simulation of the flow in a shrouded tail rotor for aero-acoustic prediction purpose. Instrumentation of a flight test helicopter (Dauphin) that will be used in a near future to validate the numerical approach is shortly described. Emphasis is laid on the implementation of CFD tools and aerodynamic prediction results. Noise prediction results are also presented. It is shown that a coupled CFD-CAA analysis can be useful to improve our physical understanding of the aero-acoustic behaviour of a Fenestron.

INTRODUCTION

In response to the need of reliable numerical tools for quiet helicopter studies, a lot of progress has been achieved in predicting main rotor impulsive noise. As regards to High Speed Impulsive noise simulation, CFD combined with integral methods has proven efficient. In the field of Blade-Vortex Interaction noise where CFD still has to solve the numerical wake diffusion problem, computational tools based on physical models have made good job in predicting and quantifying the effectiveness of noise reduction concepts.

However, the main rotor is not the only noise source of a helicopter. The tail rotor plays an important part in the helicopter acoustic nuisance so that, for instance, designing a low noise flight procedure for the main rotor does not necessarily imply low tail rotor noise. The question of tail rotor noise prediction thus arises.

Noise prediction for a shrouded tail rotor like the Eurocopter Fenestron cannot be solved by using physical models because of the complex flow field in which the rotor blades operate. In this case, CFD is required for providing input data to aero-acoustic integral methods.

To address the problem of Fenestron noise, ONERA, with the support of the French Ministry of Civil Aviation (DGAC), has launched a numerical and experimental program. The numerical aerodynamic and acoustic approach combines unsteady Navier Stokes *elsA*^{1,2} computations and

acoustic predictions using the code PARIS³ based on the Ffowcs Williams-Hawkings equation. The experimental part consists in the instrumentation and flight tests of a research helicopter.

The paper addresses the numerical aerodynamic and aero-acoustic simulations performed in the first phase of the above-mentioned program. It is at our knowledge the first study ever published on this subject. The unsteady computations performed account for interactions between tail rotor blades and hub support struts and for the complex aerodynamic inflow resulting from the presence of the fuselage and from the main rotor influence.

The paper first presents the Dauphin 6075 research helicopter on which our numerical simulations are made. The helicopter instrumentation, presently in progress, is briefly described.

Then the codes are presented together with the numerical strategy implemented in *elsA*. Three different flight cases (level flight at full and reduced loading, climb) are considered. The convergence of CFD computations is documented. Aerodynamic and acoustic results are presented and discussed. Correlation between the Fenestron blade loads and acoustics is made with emphasis on the influence of blade-to-blade acoustic interferences on the radiated noise.

In conclusion, improvements in the numerical approach for more complete account of the complexity of the problem, to be validated by future flight tests, are proposed.

THE DAUPHIN 6075 RESEARCH HELICOPTER

The Fenestron under study equips the Dauphin 6075 helicopter (Figure 1) of Istres Flight Test Center (CEV). It has 13 equally spaced blades and the hub is supported by three arms. One of these arms, bulkier than the two other ones, contains the drive shaft of the tail rotor (Figure 2). Although it corresponds to a first generation Fenestron not representative of modern ones with uneven blade spacing for acoustic concerns, this simpler design was thought to be an advantage (in a first step) for numerical simulation and physical insight.



Figure 1. The CEV Dauphin 6075 in flight.

In order to constitute an experimental database for understanding of aerodynamic and acoustic phenomena and for code validation, this helicopter is presently being instrumented in partnership between ONERA, Eurocopter and CEV. The instrumentation comprises aerodynamic and acoustic sensors.

For flow characterization, the rear part of the helicopter fuselage is fitted with 33 steady pressure sensors. In order to get the tail rotor radial velocity distribution, the Fenestron duct is equipped with 3 equally spaced radial rakes, each carrying 5 Pitot probes plus one 7-hole probe. Furthermore, 4 generating lines of the duct are fitted each with 20 steady pressure sensors and 2 unsteady Kulites, all flush mounted. Embarked microphones are also installed on the rear horizontal empennage.

Tail rotor instrumentation is as follows. Five pairs of upper/lower surface thin layer unsteady Kulite pressure transducers are distributed on 4 of the Fenestron blades. The tail rotor is also equipped with control strain gauges. Kulite and strain gauge signals together with a signal delivered by an encoder synchronized with the rotation will be acquired and conditioned in the rotating frame. They will be transmitted by radio to the cabin. A common time base for measurements performed in the rotating and non-rotating frames will be used.

Because of mechanical constraints and safety requirements related to the high rotation rate (about 5000 rpm) and to the small blade thickness (4 millimeters) and because of the small room available in the Fenestron hub for the conditioning and acquisition system, the design and realization of the instrumentation are a real challenge.

Two series of flight tests are planned at CEV, respectively in 2006 and 2007. In the first ones, detailed on-board aerodynamic and acoustic measurements will be realized. In the second ones, a deeper investigation of selected flight cases will be made using on-board aerodynamic/acoustic measurements, as previously, together with ground noise microphone measurements.



Figure 2. Starboard view of the Fenestron.

NUMERICAL INVESTIGATIONS

Simulation cases

The three configurations computed are presented in Table 1. The first configuration corresponds to a level flight at 150 knots. In the second case, the influence of a reduction of the Fenestron thrust is investigated. The last case corresponds to a high rate climb flight condition, with maximum loads on the main rotor and high thrust on the Fenestron rotor to counter the torque of the main rotor.

Table 1. Table of test cases.

		Flight speed
Case 1	Level flight	150 knots
Case 2	Level flight low thrust	150 knots
Case 3	High rate climb	75 knots

Methods used

Aerodynamics solver general description

The *elsA*¹ solver, developed by ONERA, is used to perform the different computations of the present work. This multi-application object oriented aerodynamic code solves the Euler or the Reynolds Averaged Navier-Stokes equations. It is based on a cell-centered finite volume formulation for multiblock structured meshes. It allows to simulate a wide range of aerospace configurations such as aircraft, space launchers, turbo machinery, missile configurations and helicopters.

The space discretization is based on the classical 2nd order centered Jameson's scheme employed with a scalar artificial viscosity and Martinelli's correction.

Several time discretization schemes are available to perform the unsteady computations. Explicit or implicit schemes, such as a pseudo-time approach (Dual Time Stepping) or the Gear method, are available. The time integration can be solved either by an implicit residual smoothing phase with a 4th order Runge-Kutta technique or by an implicit LU scalar relaxation phase associated to a backward Euler scheme. In the present paper, the Gear method is used to perform all the unsteady computations.

The *elsA* solver contains many turbulence models. The turbulent computations presented in this paper use the two equations Wilcox $k-\omega$ model with the Zheng limiter and correction of Kok.

The simulation of a rotating machine (propeller, rotor, ...) can be simplified by modelling the rotor with a lifting surface, called "actuator disk". It represents the time-averaged loads of the rotor in a steady flow computation. Due to the steady-state assumption, a large reduction of computational cost is achieved by comparison with an unsteady computation of the flow around rotating blades. The data for the actuator disk are generated by the Eurocopter HOST software⁴.

The Chimera method simplifies the process of structured mesh generation for complex geometry and relative moving bodies. It allows the computation of meshes made of overset grids. The Chimera² technique consists in an overlapping boundary condition and a masking condition for areas corresponding to a solid body. For both types of boundaries, the transfer of conservative and turbulent variables is carried out by interpolation. Two algorithms are implemented to solve the interpolation cell search procedure for moving or fixed meshes. The two preconditioning methods are based on a Cartesian grid or an Alternative Digital Tree. In the present computations, the Cartesian grid technique was used.

Grid topology and computation details

The numerical study has been done in two successive phases: first a steady simulation and then an unsteady simulation.

All the grids have been generated with the ICEM-CFD software. The CAD geometry (Figure 3) has been simplified (lateral stabilizer removed, simplification of the engines fairing geometry, no modelization of the rotor hub).

For the steady computations, we used two grid parts, the fuselage grid, which is also the background grid, and the main rotor actuator disk grid. The main

rotor is represented by a non-uniform actuator disk included in a two cylindrical blocks mesh with about 17,000 cells. The integration of the two main rotor grid blocks in the fuselage blocking is presented in Figure 4.

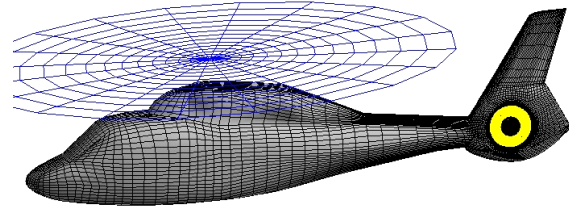


Figure 3. Skin mesh and actuator disk plans.

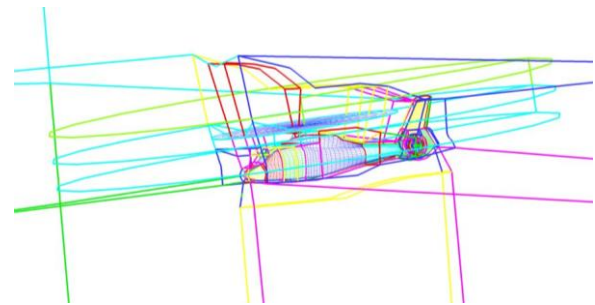


Figure 4. Skin mesh and main rotor and fuselage blocking.

The Navier-Stokes background grid contains 87 blocks and around 1.2 million of cells. If we compare with previous ONERA fuselage computation⁵ the present grid in the main part of the fuselage corresponds to a coarse grid. Indeed, it has been decided to limit the number of cells outside the Fenestron vein, for CPU time consumption reasons. This Fenestron vein grid presented in Figure 5 is more refined compared to the other grid areas, and contains for the steady case the actuator disk Fenestron rotor mesh (in yellow). The actuator disk is located at the interface of four semi cylindrical blocks. One can notice the use of the patch grid technique (one cell over four) in order to have a well adapted azimuthal refinement. This part of the grid contains about 400,000 cells. The three supports of the Fenestron hub have also been taken into account.

For the unsteady computations the background grid is the same as for the steady cases. The only modification is the merge of the four blocks of the actuator disk surface into only two semi cylindrical blocks. The unsteady Chimera interpolation research of the thirteen blades of the Fenestron rotor is limited to these two very refined blocks (1 degree azimuthally spaced as shown in orange and green planes in Figure 6). This discretization is necessary to avoid critical points in the interpolation search procedure. For unsteady computations, the

interpolation search has to be made at each time step. Note that there is no Fenestron blade tip leakage modeled, to reduce the total number of grid points.

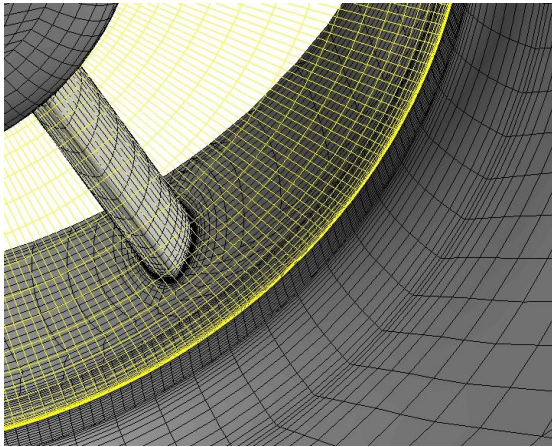


Figure 5. Zoom in Fenestron vein grid.

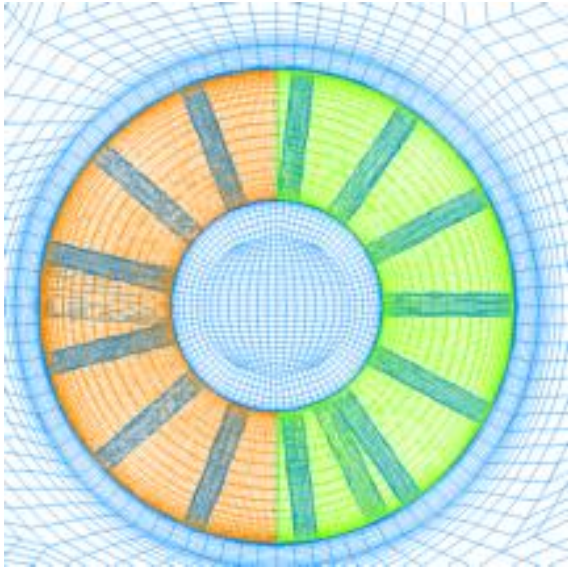


Figure 6. Fenestron grids details for unsteady calculations.

Each of the 13 Fenestron blades is meshed with two blocks, which contain about 90,000 cells. This grid is duplicated for each blade with successive rotation transformations by $2\pi/13$. As shown in Figure 7, each blade mesh remains contained inside the two blocks with higher density of the background grid. Each blade grid only interacts with the background grid. There is no interaction between the blade grids themselves in order to reduce cpu time significantly.

The complete unsteady grids system has got 2.3 million cells.

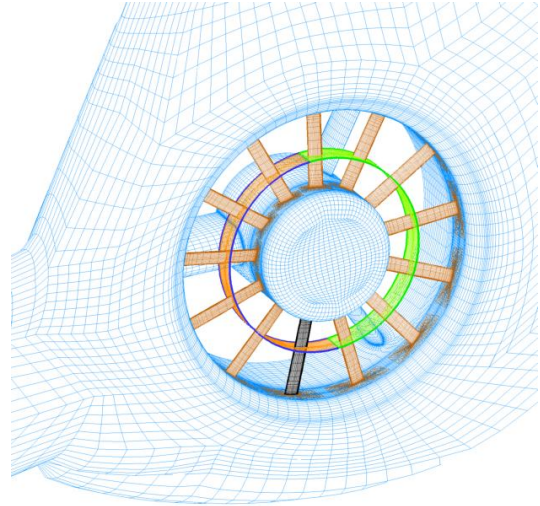


Figure 7. Fenestron blade Chimera details.

The unsteady blade pressures were extracted from the aerodynamic simulation at each time step during a final computation over an azimuthal angle equal to $2\pi/13$, in order to have the 360 degrees evolution on the blade surface of the Fenestron rotor. These pressures are then transmitted to the acoustic software PARIS.

Acoustic solver description

The acoustic predictions presented below have been performed using the acoustic code PARIS which computes the loading and thickness noise radiation, based on the Ffowcs Williams-Hawkings equation and a time-domain formulation designed at ONERA. Input data for PARIS are the blade geometry and kinematics and the steady or unsteady blade pressure distributions extracted from an *elsA* computation. Noise calculations for a 13 blade rotor are performed assuming a uniform inflow. No influence of solid surfaces, nor of the inhomogeneous inflow (accounted for in the CFD blade pressure prediction) is accounted for in the noise radiation computation. Furthermore, only discrete frequency noise is computed.

Convergence of CFD simulations

Steady results

For the steady aerodynamic simulations, 6000 iterations were performed, resulting in a satisfying reduction of the mean residuals by more than 4 orders.

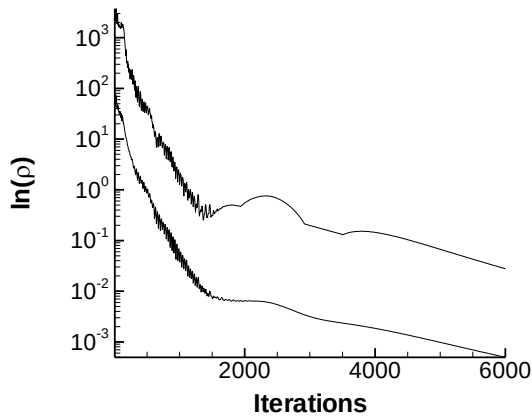


Figure 8. History of density residuals.

The Yplus contours drawn in Figure 9 present values below five which is in accordance with previous studies for a coarse grid on the fuselage. Higher values are obtained on the trailing edge of the vertical stabilizer, due to insufficient grid density in this part of the fuselage, with no consequence for the present study, which aims at analyzing the flow field in the Fenestron vein.

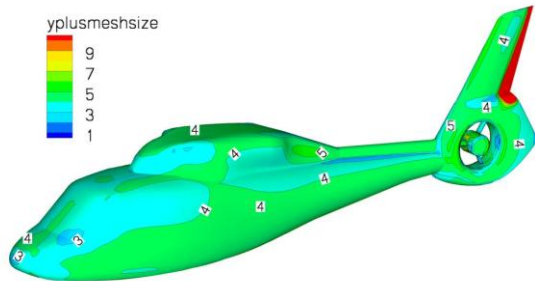


Figure 9. Steady case Yplus contours.

Unsteady results

For unsteady computations, the Gear time integration technique was used with an azimuthal step equal to 1 degree and a sub-iteration absolute residual tolerance of 10^{-6} . By convention, the blade azimuth is 0 degree when the Fenestron blade is up (Figure 10, left). The tail rotor rotates clockwise when looking from the Fenestron inlet. At iteration 0 Blade 1 (reference blade) is set to Azimuth -90 degrees (Figure 10, right). The blades are numbered clockwise.

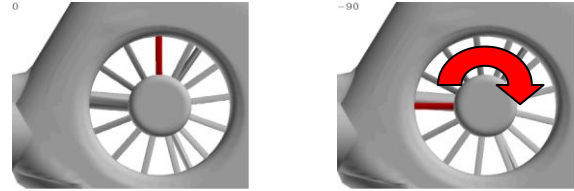


Figure 10. Position of blades for 0 and -90 degrees.

Three complete revolutions of the Fenestron rotor were necessary to obtain the converge of the rotor thrust (Figure 11). The cpu time is around 90 hours on one processor of a NEC-SX6.

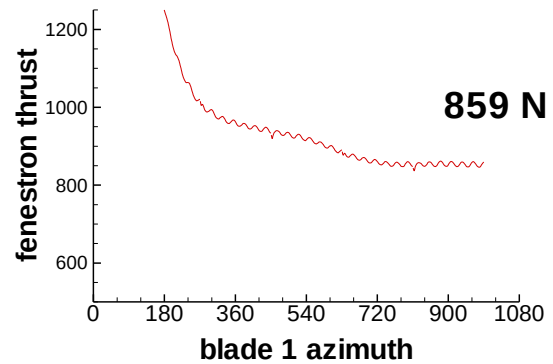


Figure 11. Fenestron thrust history.

The individual blade thrust history is plotted in Figure 12, where each blade thrust is almost similar to the next blade shifted by $360/13$ iterations. The thrust convergence is clearer if we use a shifted iteration for the X-axis corresponding for each blade to its own iteration position (Figure 13). We can see that the convergence is almost reached, with minor evolutions after the shifted iteration 1080 (3 revolutions).

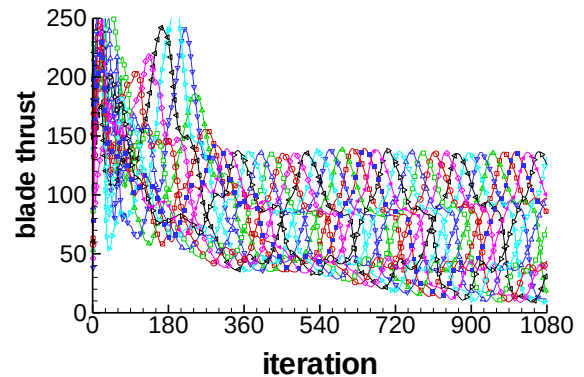


Figure 12. Individual blades thrust history versus iterations.

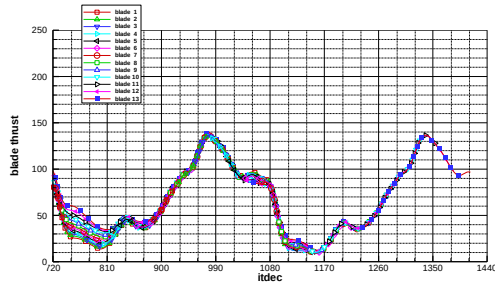


Figure 13. Blades thrust history versus shifted blade iteration.

Discussion of results

Steady results (aerodynamics only)

The steady results presented hereafter are only for the level flight at 150 knots.

The pressure distribution (Figure 14) shows the location of the stagnation point on the nose of the fuselage for this configuration. We can also notice the diminution of the pressure in the part of the Fenestron vein located upstream and the increase in the part of the vein located downstream. We can also see the stagnation area on the hub support, as well as on the leading edge of each of the three stator blades.

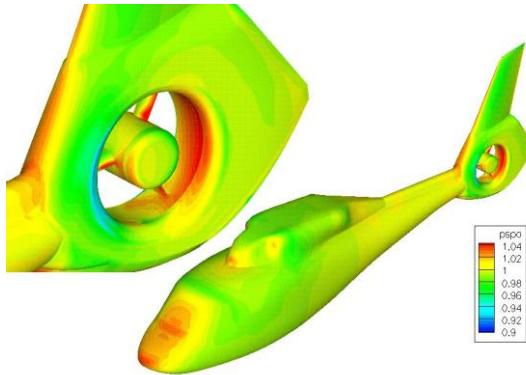


Figure 14. Steady results Case 1. Pressure distribution on the fuselage.

The influence of the uniform actuator disk can be clearly identified in a horizontal cut of the Fenestron at mid-vein, resulting in a sharp increase of the total pressure ratio at the actuator disk plane location (Figure 15). Looking at the surface restricted streamlines, several recirculation area can be noticed: the first one is located at the upwind entrance of the vein (just in front of the biggest hub support) and the second one with less extension just after the hub. Note a third zone of recirculating flow as a consequence of flow separation after the trailing edge of the hub support.

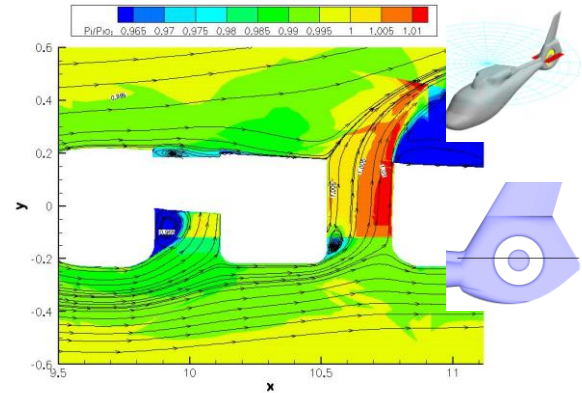


Figure 15. Steady results Case 1. Streamlines and total pressure levels in the Fenestron at mid-vein.

A close view at the 3D streamlines in the Fenestron vein (Figure 16) confirms the strong influence of the biggest hub support, and the totally inhomogeneous flow field in the vein.

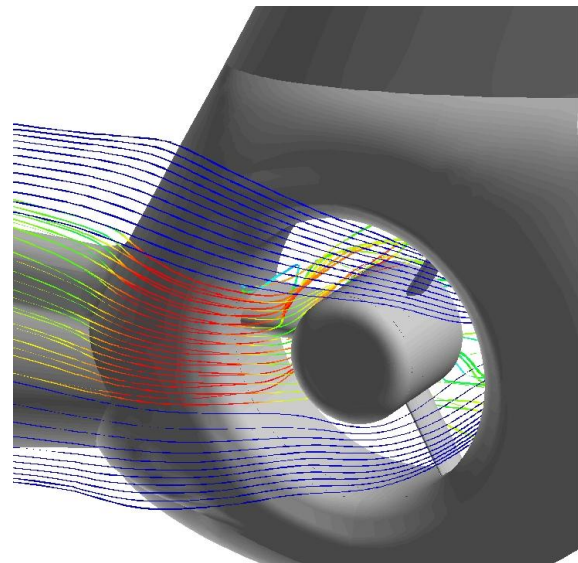


Figure 16. Steady results Case 1. Detailed view of streamlines in the Fenestron.

Unsteady results (aerodynamics + acoustics)

Case 1 (level flight, 150 knots)

The pressure distribution on the fuselage (Figure 17) is globally comparable to the steady one (Figure 14). Unsteady effects are negligible outside the Fenestron vein. The incoming Fenestron streamlines (Figure 17) do not pass through the main rotor disk. In the Fenestron vein, the streamlines drawn at a given azimuth show also the large impact of the biggest hub support (Figure 18).

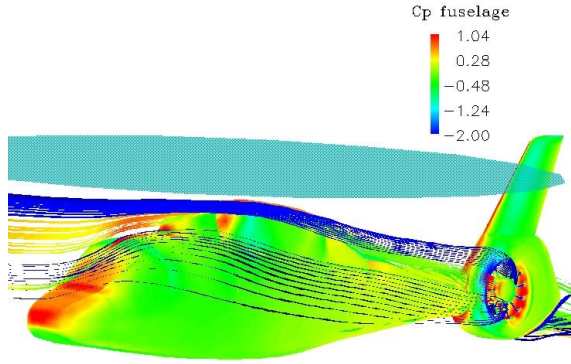


Figure 17. Case 1. Streamlines coming through the Fenestron and pressure distribution.

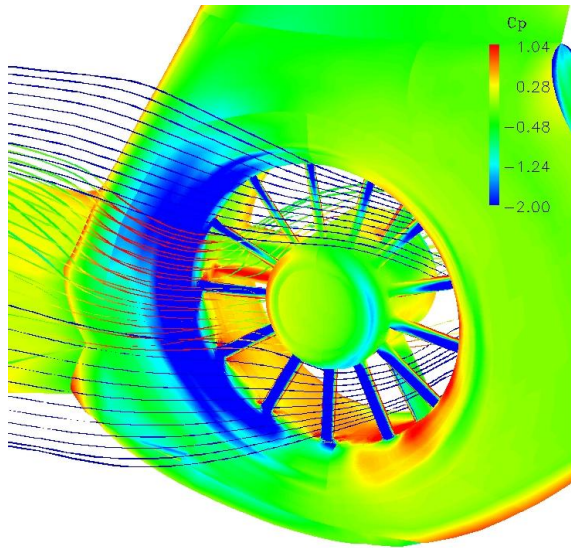


Figure 18. Case 1. Detailed view of streamlines in the Fenestron and pressure distribution.

The pressure distribution in the vein has been plotted for one Fenestron reference blade position in Figure 19 (the pressure is maximum in red and minimum in blue). We can see the low pressure zone on the blade suction sides following the evolution of the blade sectional Mach number and the perturbation of the pressure on the fairing due to the blade tip. We can also notice that the pressure levels are lower on the suction side of the blade when the blade is down (which is a kind of advancing blade) and higher when the blade is up (retreating blade).

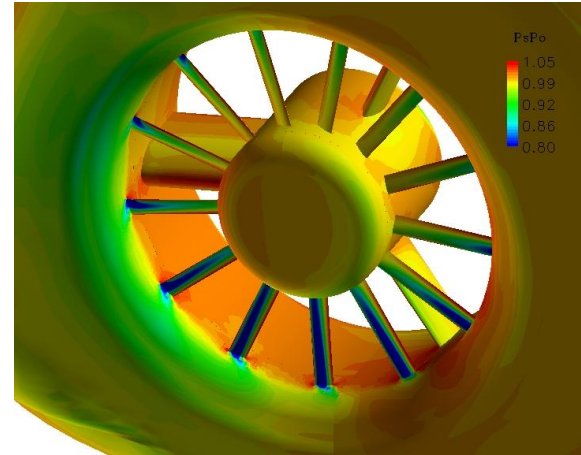


Figure 19. Case 1. Pressure in the Fenestron vein, blade 1 azimuth -90.

The evolution of blade thrust is presented in Figure 20 as a function of azimuth, for one revolution. We can notice that the blade thrust is the lowest at Azimuth -30 degrees and maximum at 165 degrees; there are also local maxima around 30 degrees and 240 degrees, which can be due to the influence of the hub supports, especially near Azimuth 270 degrees. The total variation of each blade thrust is about 125 N, which is quite large compared to the mean value. We can also notice a very sharp decrease during the beginning of the retreating phase (between 270 and 300 degrees) where half of the thrust is lost.

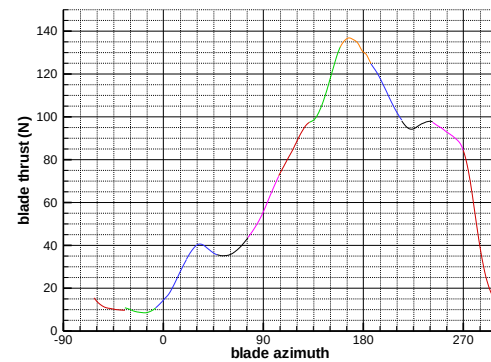


Figure 20: Case 1. Evolution of blade thrust during one revolution.

The blade loads radial distribution is presented in Figure 21 as a function of azimuth. We can notice the high variation of the loads in both azimuth and radial directions. Especially the low loads just after the big strut for azimuths comprised between -70 and 0 extend on all the blade in radial direction except on a narrow band at the blade tip.

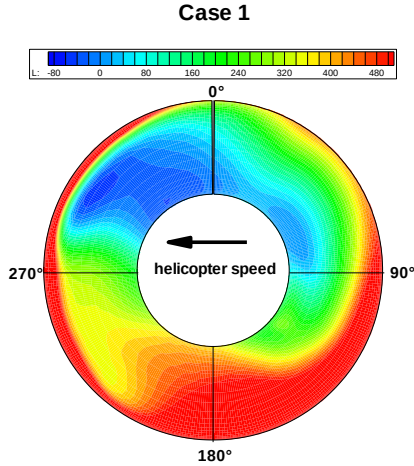


Figure 21. Azimuthal evolution and radial distribution of sectional loads. Level flight (Case 1).

Acoustic results are presented in terms of dBA noise contours. All levels are referenced to a common value. The observation plane is horizontal and located 150 m below the helicopter. Predicted levels for thickness, loading and total noise are shown in Figure 22. The rear part of the helicopter is on the right side of the maps and the inflow through the Fenestron inflow is going from bottom to top. The Fenestron is in a plane perpendicular to the x-axis of the figures in (0,0.).

The maximum thickness noise level has been found lower than the maximum loading noise one by 10 dBA. Total noise is thus widely dominated by loading noise and only loading noise level will be presented in the following sections.

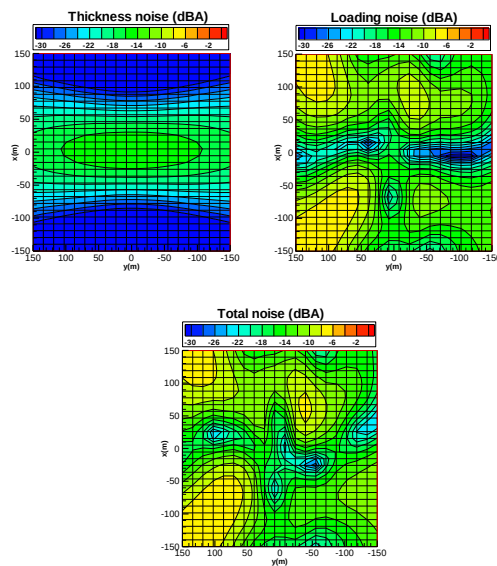


Figure 22. Case 1. Predicted dBA noise level in a horizontal plane 150 m below the helicopter.

Case 2. (Thrust effect in level flight)

In practice the thrust is reduced by 50% compared to Case 1, by decreasing the blade pitch angle by 6 degrees.

Globally, the incoming streamlines in the Fenestron are similar to the previous case (these streamlines do not pass through the main rotor disk). In the Fenestron vein, we can notice that two of the streamlines do not enter the Fenestron vein due to the reduced thrust (Figure 23). As expected, the blade pressure distribution shows less blue zones on the blade suction sides, confirming lower loading on the blades (Figure 24).

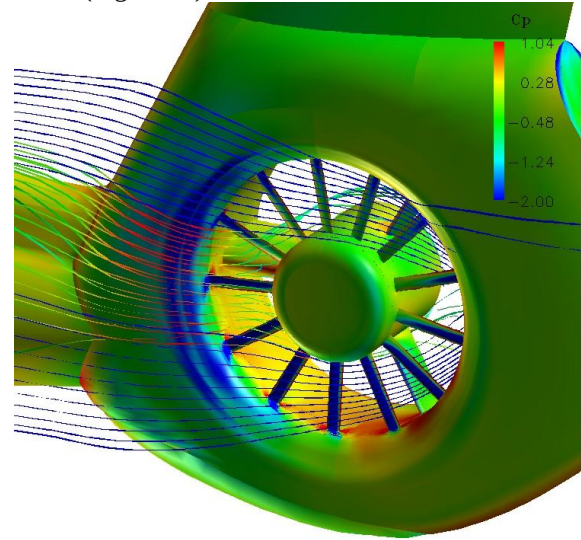


Figure 23. Case 2. Detailed view of streamlines in the Fenestron and pressure distribution.

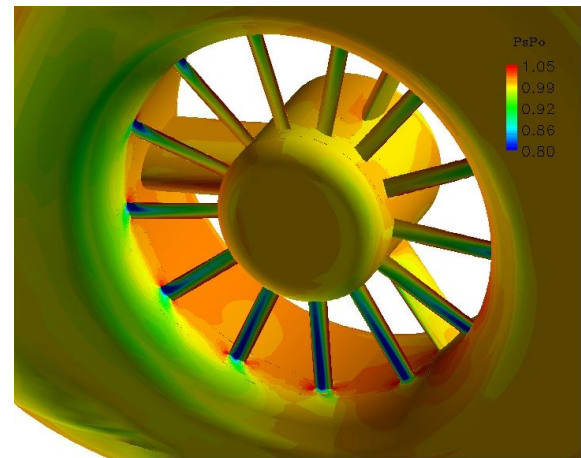


Figure 24. Case 2. Pressure in the Fenestron vein, Blade 1, Azimuth -90.

The evolutions of blade thrust as a function of azimuth for Cases 1 and 2 are compared in Figure 25, for one revolution. If we compare both cases, we can

notice opposite thrust values at Azimuth -10. The influence of the support located at Azimuth 120 degrees is more noticeable in the low thrust Case 2. The other main difference is the reduced mean thrust value and reduced peak-to-peak amplitude in Case 2.

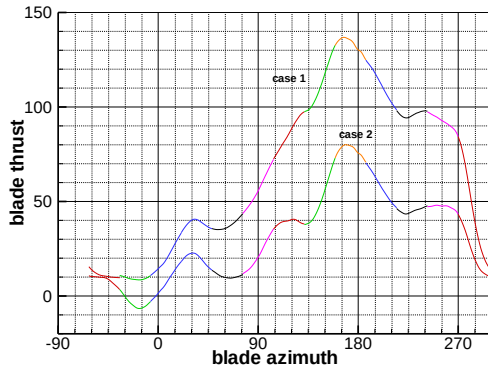


Figure 25. Cases 1 and 2. Evolution of blade thrust during one revolution.

Blade loads comparison is now presented in Figure 27 as contour plots of sectional loads. There is a global sectional loads reduction in Case 2, all over the disk. The high variation of the loads in both azimuthal and radial directions can be noticed for both cases. The low loads area just after the big strut for azimuth between -70 to 0 has a larger extension in Case 2.

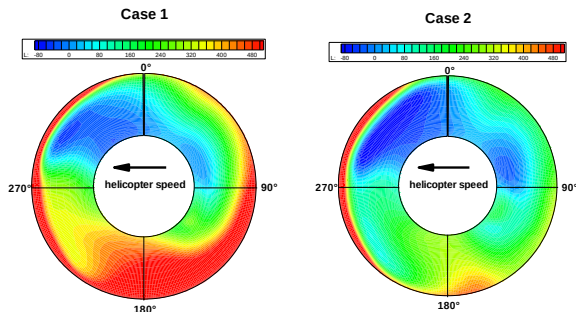


Figure 26. Azimuthal evolution and radial distribution of sectional loads. Comparison between Case 1 and Case 2 (less loaded).

Loading noise results for this case (Figure 28) are surprising at first sight. Though loads and peak-to-peak variations are weaker in Case 2 than in Case 1, particularly near the blade tip (see Figure 27), predicted loading noise is higher in Case 2: -6dBA expected from a gross estimation based on global loads, about +2dBA predicted instead.

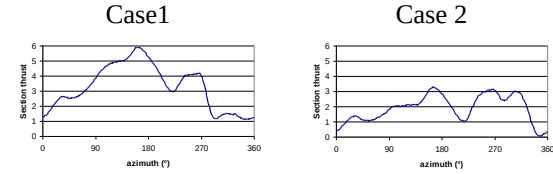


Figure 27. Azimuthal evolution of sectional loads at 0.9R. Comparison between Case 1 and Case 2.

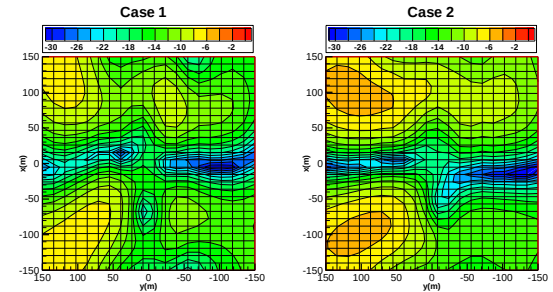


Figure 28. Predicted dBA noise level in a horizontal plane 150 m below the helicopter. Comparison between Case 1 and Case 2.

To explain this result, the noise radiated by only one blade in rotation has been calculated for a receiver which approximately corresponds to the maximum noise location in the observation plane and compared with the predicted noise resulting from the thirteen blades. Figure 29 shows that the predicted noise for the isolated blade is lower when the blade is less loaded, as expected.

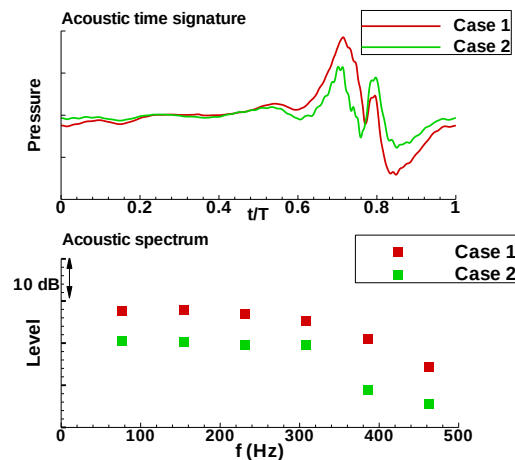


Figure 29. Acoustic time signature and spectrum of a single blade, calculated on same receiver. Comparison between Case 1 and Case 2.

In Figure 30, global noise signatures and spectra of the thirteen rotor blades are shown for Cases 1 and 2. Time signatures are displayed over $1/10^{\text{th}}$ of revolution so that they can be compared more easily.

The figure shows that, when thirteen blades are considered, noise contributions from individual blades combine in a more constructive way in Case 2 than in Case 1, providing a higher noise level.

The resulting noise thus appears not simply related to global loads and azimuthal blade loading variations but also strongly depends on the acoustic interferences between Fenestron blades.

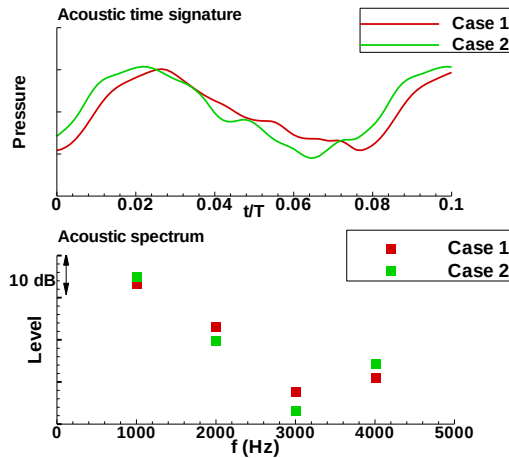


Figure 30. Global acoustic time signature and spectrum for thirteen blades, calculated on same receiver. Comparison between Case 1 and Case 2.

Case 3. (Climb case)

This case is very different from the two previous ones. First it is a high climb rate flight condition at 75 knots, second the climb slope is high (8 degrees), third the power and the thrust applied on the Fenestron are important. The blade pitch angle used is 6 degrees higher than for the level flight case. For all these reasons four revolutions are needed to obtain a periodic blade thrust convergence on this case.

The streamlines around the fuselage (Figure 31) show the particularity of this climb case, where the Fenestron is impacted by the main rotor wake. In this particular case, complete unsteady simulation for the main rotor could be interesting to quantify the influence of the main rotor on noise compared to the actuator disk modelization.

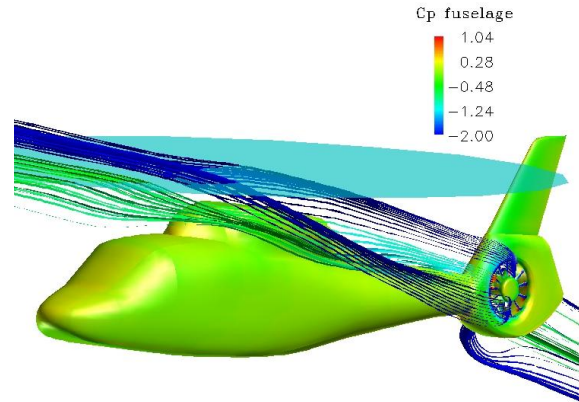


Figure 31. Case 3. Streamlines coming through the Fenestron vein and pressure distribution.

In the Fenestron vein (Figure 32), we can notice that the streamlines pass on the other side of the lower strut of the hub (comparison with Figure 18), due to the higher aspiration through the Fenestron (higher loading). The low pressure on the blade suction sides shows the high level of thrust in this case in the entire Fenestron vein (Figure 33).

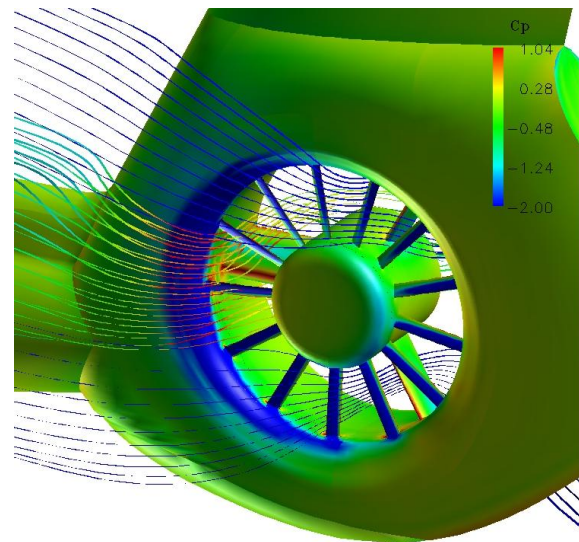


Figure 32. Climb case. Detailed view of the streamlines in the Fenestron and pressure distribution.

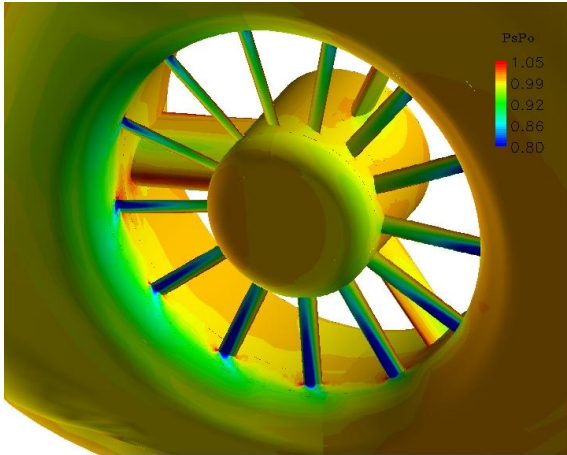


Figure 33. Climb case. Pressure in the Fenestron. Blade 1 Azimuth -90.

Considering the evolution of the blade thrust versus blade azimuth in Figure 34, we can notice a peak-to-peak thrust variation of 90 N, which is lower than for level flight case, despite the higher mean Fenestron thrust. This is probably due to the combination of reduced advancing velocity (150 knots in Case 1 and 75 knots in Case 3) which reduces the dissymmetry between advancing and retreating blades. Nevertheless the azimuthal evolution of the thrust shows minima around -30 degrees and maxima at 160 degrees and two local maxima corresponding to the big strut at 270 and the upper strut at 30 degrees. The influence of the third support (lower strut) is hardly noticeable.

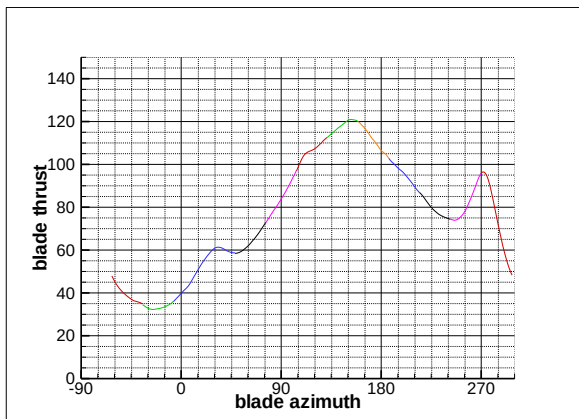


Figure 34. Case 3. Evolution of blade thrust during one revolution.

The azimuthal evolution of blade sectional loads shows lower values in the upper part of the rotor disk, and higher ones in the low part (Figure 35). The radial loading distribution is similar to Case 1 with an increase of thrust from the blade root to the tip. However, peak-to-peak sectional loads variations in

the tip region (Figure 36) are lower than for Case 1 (Figure 27).

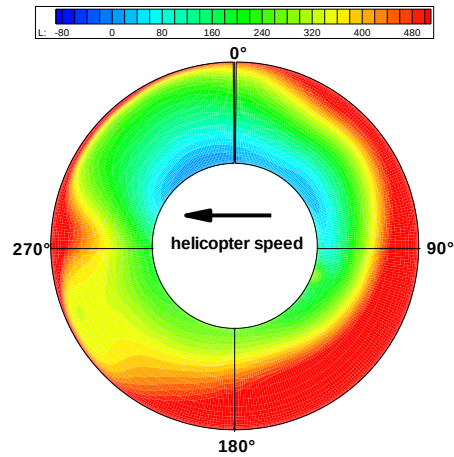


Figure 35. Azimuthal evolution and radial distribution of section load: climb case.

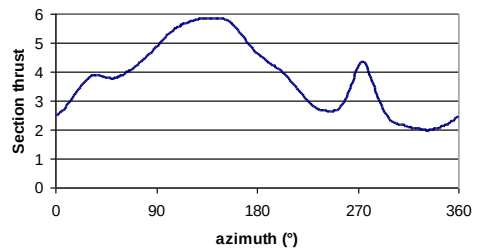


Figure 36. Azimuthal evolution of section load at 0.9R: climb case.

The acoustic results are presented on Figure 37. The comparison with Figure 28 shows that the maximum loading noise level is found lower than for Case 1 by about 1 dBA though the global thrust is higher.

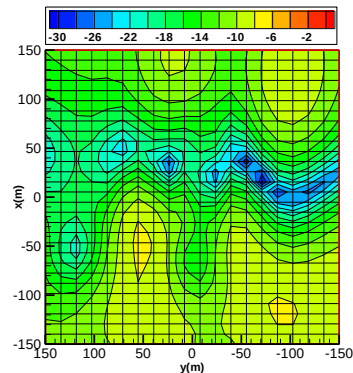


Figure 37. Predicted dBA noise level in a plane 150 m below the helicopter. Climb case.

Thus result again shows that the noise radiation strongly depends on azimuthal blade loading variations and acoustic interference between rotating blades.

CONCLUSION AND FUTURE WORK

A numerical approach aiming at evaluating the noise generated by a shrouded tail rotor (Fenestron) has been presented in this paper. Most of the effort was devoted to the choice of a numerical strategy to compute the complex aerodynamic flow field in the Fenestron vein.

A steady approach, where both the main and tail rotors are modelled by an actuator disk, has shown the heterogeneity of the inflow at the Fenestron inlet for advancing flight conditions, combined with areas of flow separation of significant extension. The hub supports (stator) significantly modify the flow-field azimuthal heterogeneity. For climb conditions, the main rotor wake can directly be ingested inside the Fenestron vein, adding complexity to the aerodynamic flow field.

Thanks to an overlapping grid strategy (Chimera technique), unsteady viscous simulations have been performed for different flight conditions. Predicted unsteady pressure fluctuations experienced by the tail rotor blades have been used as an input for the noise evaluation. Loading noise has been found to be dominant, compared to thickness noise, as a consequence of tail rotor-stator interactions, emphasized by the fact that the support enveloping the drive shaft has got a bulkier shape than the two other ones. An important result is that no direct relationship between tail rotor thrust and radiated noise. Particularly, flight conditions with lower Fenestron blade loading may lead to increased radiated noise, because of specific acoustic blade-to-blade interferences.

Experimental results are of course required in order to validate the conclusions and the accuracy of the numerical approach. This will be done thanks to flight tests of a Dauphin instrumented helicopter that will be performed in a near future. Furthermore, the numerical approach is expected to be improved. As far as aerodynamics is concerned, some parts of the Dauphin geometry that could influence the aero-acoustic results (the horizontal stabilizers for example) will be included in the calculations. An unsteady approach for the main rotor is also envisaged to quantify the main rotor unsteady effects especially for climb flight conditions. The acoustic analysis will also be complemented by accounting for acoustic reflections on the Fenestron duct and for possible volume sources in the transonic regime by

using a Ffowcs Williams-Hawkings porous surface approach.

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